Darren Cornelius Suwandi

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RESEARCH INTERESTS

Adversarial Machine Learning, Explainable AI (XAI), Behavioral Threat Detection, LLM Engineering, AI-Driven Full-Stack Systems.

EDUCATION

Xiamen University Malaysia Sep. 2025 – Expected Aug. 2028

Bachelor of Digital Media and Technology | *CGPA: 3.86/4.00* | *Ranked #2 in Major* Selangor, Malaysia

- **Core Coursework**: Data Structure, Algorithm Analysis and Design, Software Architecture and Design Patterns, Operating Systems, Database, Principle of Artificial Intelligence, Big Data Analytics, Information Security, Software Project Management

RESEARCH EXPERIENCE

Xiamen University Malaysia May 2026 – Present

Undergraduate Researcher (Supervised by Ts. Dr. Teh Jia Yew and Dr. Goh Sim Kuan) Sepang, Malaysia

- Study adversarial robustness and XAI for 27-class Windows PE malware classification using 21,617 sandbox samples and 11 registry, process, file, API/DLL, and network-behavior counters.
- Compare **XGBoost** defensesincluding SHAP-guided and adversarial augmentation, monotone-constrained detection, and moving-target routingagainst add-only feature-inflation attacks.
- Use hash-grouped evaluation across 5 split seeds and 3 training seeds; adversarial augmentation outperformed SHAP-guided augmentation by 0.23–0.25 macro-F1 and 0.23–0.24 TPR at 5% FPR across four attacks, with Holm-significant results and gains of 0.19/0.21 on a held-out 10,000-query differential-evolution attack.

FEATURED PROJECTS

Adept | *Next.js, FastAPI, PostgreSQL (pgvector), RunPod, Gemini API* Oct. 2026 – Nov. 2026

- Led system design and AI engineering of a career-matching and reskilling platform within a 4-person team (**Top 7 of 213 teams**), ranking future roles with explainable skills-gap scores under moderate and fast AI-disruption scenarios.
- Built **arXiv**, **BLS**, and **Talent Corp** scraping pipelines and **MiniLM** embeddings feeding a trend-analysis and matching engine, with LLM outputs restricted to validated catalogue IDs and verified by **940+** automated tests.

Drift Anchor: Network Intrusion Pipeline | *Python, XGBoost, Zeek, AWS SQS, scikit-learn* Oct. 2026 – Present

- Engineered an **XGBoost** multiclass anomaly detection model on the UNSW-NB15 dataset, architecting a rigorous data auditing pipeline that purged 133,000+ leaked train/test pairs and 94,000+ duplicates to guarantee zero-leakage evaluation.
- Executed a 6-level feature ablation study, achieving an 86.4% accuracy and 0.8680 Weighted F1 score while actively isolating load-bearing sliding-window signals from artificial simulation artifacts to ensure real-world generalization.
- Developed a stateful, dependency-free **Zeek** log ingestion layer utilizing memory-efficient sliding deques for real-time feature reconstruction, reliably routing live network traffic to an environment-switched **AWS SQS** publisher.

Whatlf Forecaster | *Next.js, FastAPI, FLAML, SHAP, Scikit-learn* Apr. 2026 – Present

- Engineered a decoupled AutoML SaaS platform utilizing **FastAPI** and **FLAML** to automate model selection (e.g., Random Forest) with real-time training logs streamed via Server-Sent Events (SSE).
- Developed a highly reactive **Next.js** and Zustand dashboard featuring interactive forecasting simulations and **SHAP** integrations for local feature influence analysis.

Logic Node: MITRE ATT&CK Extraction | *PyTorch, QLoRA, FAISS, Weights & Biases* Sep. 2026 – Oct. 2026

- Engineered an offline machine learning pipeline fine-tuning a cybersecurity-domain **LLaMA-3.1** model via **QLoRA** (4-bit NF4 quantization) to extract structured threat intelligence within a strict 8GB VRAM budget.
- Architected a retrieval-augmented training pipeline utilizing **FAISS** and BGE embeddings to inject contextual technique candidates across a 19,578-sample dataset, implementing loss masking for assistant-only response optimization.
- Evaluated training dynamics via **Weights & Biases**, achieving a 0.2428 validation loss, 99.7% JSON validity, and an 83.97% exact match rate on a held-out corpus before deploying production weights to **Hugging Face**.

Resume Forge | *Next.js, FastAPI, Gemini API, Supabase, Redis* Feb. 2026 – Aug. 2026

- Architected a dual-phase ATS scoring engine and fault-tolerant Gemini API integration layer, blending deterministic Python extraction with strict LLM auditing for recruiter-aligned feedback.
- Developed an agentic editing suite featuring an SSE streaming chatbot and skill-gap analyzer, leveraging Zustand state management to instantly apply structured Before/After rewrite blocks.
- Implemented a comprehensive tiered credit system utilizing Redis for guest session tracking, Stripe for payment processing, and Supabase JWTs for secure API endpoint authentication.

Omni-QC SMT Quality Control System | *React, XGBoost, SHAP, IPC-2591 CFX* Mar. 2026 – May 2026

- Directed a 7-person engineering team as Project Lead and Frontend Architect, managing a 3-person interdisciplinary advisory board to deliver a predictive defect-routing SaaS, reducing total AOI inspection workloads by 30–50% via real-time IPC-2591 CFX machine data evaluation.
- Engineered a high-performance **React** and **TypeScript** dashboard that bridges complex **XGBoost** and **SHAP** anomaly detection models with transparent, human-in-the-loop prescriptive recommendations for factory technicians.

KerjaCerdas Job-Matching Platform | *FastAPI, LangGraph, React, pgvector, Gemini API* Apr. 2026 – Jun. 2026

- As Product/UX Lead, architected a **LangGraph** ReAct Supervisor Swarm routing parallel worker agents for resume parsing, skill-gap analysis, and career advisory, alongside an E-KYC verification pipeline.
- Built a hybrid ranking engine (cosine, skill, region, salary fit) on **PostgreSQL (pgvector)** with Gemini embeddings, exposing 5-component explainability scores through a reactive **React/Zustand** dashboard.

AskA Maintenance Automation | *n8n, LLM, MySQL, Google Workspace APIs* Mar. 2026 – Apr. 2026

- Collaborated in a 5-person team with Data Science and Electrical and Electronics Engineering (EEE) students, commissioned by Angie Ng (MQA - IT Services), to automate XMUM's maintenance infrastructure via **n8n**.
- Engineered intelligent email routing and automated worker assignment via Google Calendar while deploying an LLM conversational agent to triage requests and optimize workflow efficiency.

HONORS & AWARDS

School of Computing and Data Science Dean's List Award, Xiamen University Malaysia 2024, 2025, 2026

Full Tuition Scholarship, Xiamen University Malaysia 2024 – Present

Silver Award, CICSIC SEA (Omni-QC, Undergraduate Division) 2026

Finalist, Digdaya (KerjaCerdas), SDC Hackathon (Adept) 2026

EXTRACURRICULARS & COMPETITIONS

Active Hackathon Competitor 2026

- Competed in SEA-CICSIC 2026, Digdaya 2026, UM Hackathon 2026, and MyHack Google Hackathon 2026
- Engineered rapid-viability full-stack AI prototypes under strict 24-to-48-hour competitive constraints.

Artificial Intelligence Club | *Member* Sep. 2025 – Present

- Participated in technical discourse and collaborative initiatives focused on Git workflows, n8n automation, machine learning, and deep learning.

Coding Club | *Member* Apr. 2026 – Present

- Engaged in practical engineering workshops focusing on CI/CD pipelines and full-stack web development.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, SQL (MySQL)
Tools & DevOps: n8n, Unity, Blender, Android Studio, Wireshark, Weka
Frameworks: Next.js, FastAPI, PyTorch, Scikit-Learn, Optuna, SHAP, SMOTE, Pandas, Pydantic